

Discrete Mathematics

1001

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第一章 Logic and Proofs

The Only Truth is the Truth Table.

1.1 Basic

Proposition: A declarative sentence that is either True or False, but not both.

not declarative: $x + 1 = 3$

not T or F: Paradox. e.g. This sentence is false.

allow its truth value be undetermined: He will go to Paris.

Proposition Variable: p, q, r , etc.

Atomic Proposition: the simplest proposition.

Given proposition $p \rightarrow q$:

Converse 逆命题	$q \rightarrow p$
Inverse 否命题	$\neg p \rightarrow \neg q$
Contrapositive 逆否命题	$\neg q \rightarrow \neg p$

p : **Hypothesis** / **Antecedent** / **Premise**

q : **Conclusion** / **Consequence**

Tautology: always True;

Contradiction: always False;

Contingency: sometimes True, sometimes False;

Satisfiable: 存在赋值方式使为真, Tautology + Contingency

Precedence: $\forall/\exists > \neg > \wedge/\vee > \rightarrow / \leftrightarrow$

Theorem: a somewhat important statement.

Proposition / **Facts** / **Results:** less important.

Axiom 公理 / **Postulate** 假设: can be used as premises.

Lemma: a less important theorem that is helpful for other results.

Corollary 推论: a theorem that can be established directly from a theorem.

Conjecture 猜想: guess to be true.

1.2 Logical Operators

1.2.1 Defines

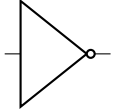
Negation 否定	$\neg x / !x / \sim x$	not		$1 - x$
Conjunction 合取	$x \wedge y$	and		xy
Disjunction 析取	$x \vee y$	or		$x + y - xy$
Exclusive Or	$x \oplus y$	xor		
Conditional / Implication 蕴涵	$x \rightarrow y$	if, then	$\neg x \vee y$	$1 - x + xy$
Biconditional / Equivalence 等价	$x \leftrightarrow y$	iff	$\neg(x \oplus y)$	

x only if y: $x \rightarrow y$

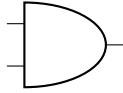
x unless y: $y \rightarrow \neg x$

x, y both occur: $x \wedge y$, no causality.

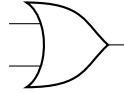
avoid using *any*, 因为自然语言下同时有任意一个/某一个的含义。



Inverter



And Gate



Or Gate

1.2.2 Rules

Logical Equivalence:

- Identity: $x \wedge \mathbf{T} \equiv x$, $x \vee \mathbf{F} \equiv x$
- Domination: $x \wedge \mathbf{F} \equiv \mathbf{F}$, $x \vee \mathbf{T} \equiv \mathbf{T}$
- Idempotent: $x \wedge / \vee x \equiv x$
- Double Negation: $\neg(\neg x) \equiv x$
- Communtative & Associative Laws: $\wedge/\vee$ 分别满足交换 & 结合律
- Distribution: $\wedge/\vee$ 分别对 $\vee/\wedge$ 分配

$$\begin{cases} x \vee (y \wedge z) & \equiv (x \vee y) \wedge (x \vee z) \\ x \wedge (y \vee z) & \equiv (x \wedge y) \vee (x \wedge z) \end{cases}$$
- De Morgan's Laws: flip operator

$$\begin{cases} \neg(x \wedge y) & \equiv \neg x \vee \neg y \\ \neg(x \vee y) & \equiv \neg x \wedge \neg y \end{cases}$$
- Absorption Laws:

$$\begin{cases} x \vee (x \wedge y) & \equiv x \\ x \wedge (x \vee y) & \equiv x \end{cases}$$

1.2.3 Normal Forms

Simple Conjunctive/Disjunctive Formula: only consists of $\wedge/\vee$ and $\neg$

Conjunctive/Disjunctive Normal Form (CNF / DNF): $\bigwedge(\vee) / \bigvee(\wedge)$

主合取/析取范式: 外层为合取/析取。CNF / DNF 必存在, 构造性证明, 略。

CNF / DNF 不唯一, 故引出 **Principle/Major C/D NF** (P*NF/M*NF) (主合/析取范式), 强制约定每个基本项包含所有 n 个变元。P*NF 是唯一的。

极小项: $\bigwedge_{i=1}^n m_0 \sim m_{2^n-1}$;

极大项: $\bigvee_{i=1}^n M_0 \sim M_{2^n-1}$;

编号: from left to right, i -th bit = $[p_i]$ 。

e.g. $(p_1 \wedge \neg p_2 \wedge \neg p_3) = m_{100} = m_4$; $(\neg p_1 \vee p_2) = M_{01} = M_1$

Construct P*NF base on *NF: for each simple formula with less than n variables, adjust, and merge the same formulas

a. $A \equiv (A \wedge p_k) \vee (A \wedge \neg p_k)$

b. $A \equiv (A \vee p_k) \wedge (A \vee \neg p_k)$

转换 PCNF/PDNF, 其中 $\sim$ 为按位取反: $\bigwedge_{t \in S} M_t = \bigvee_{t \in U \setminus S} m_{\sim t}$; $\bigvee_{t \in S} m_t = \bigwedge_{t \in U \setminus S} M_{\sim t}$

证明: PDNF $\rightarrow$ PCNF: 每个 $t \in S$ 表示一种成真赋值, 则每个 $t \in U \setminus S$ 表示一种成假赋值。为使结果为真, 则每种成假赋值均需有一项不成立, 即补集的按位取反。易知该操作可逆, 则 PCNF $\rightarrow$ PDNF 亦得。

1.3 Predicates and Quantifiers

1.3.1 Defines

Predicate 谓词: refer to a property, propositional function $P, P(x_1, \dots, x_n)$

Quantifier 量词: the extent to which P is true over a range of elements

a. universal quantifier 全称量词: $\forall$

b. existential quantifier 存在量词: $\exists$

c. uniqueness quantifier 唯一量词: $\exists!/ \exists_1$

Domain: the domain of x ; the domain consists of $a, b, c, \dots$

Scope: the scope of $\forall/\exists$, etc.

Bound Occurance 约束出现: x is restricted by quantifiers.

Free Occurance 自由出现: x is not restricted by any quantifiers.

Closed Form 闭式: no free variables, all variables is in some scope.

1.3.2 Rules

- Negation:

a. $\neg \forall \equiv \exists \neg$

b. $\neg \exists \equiv \forall \neg$

- Distribution: turn $\rightarrow / \leftrightarrow$ to $\wedge / \vee$ first
 - a. under the same restriction: only the following two are correct
 - $\forall x(A(x) \wedge B(x)) \equiv \forall xA(x) \wedge \forall xB(x)$
 - $\exists x(A(x) \vee B(x)) \equiv \exists xA(x) \vee \exists xB(x)$
 - b. under different restriction: $\forall/\exists x(A(x) \wedge / \vee B) \equiv \forall/\exists xA(x) \wedge / \vee B$

1.3.3 Prenex Normal Form

前束范式: all quantifier are at the beginning, and their scopes last to the end.

e.g. $\forall x \exists y \forall z (P(x, y) \vee Q(x, y, z) \vee \neg R(z))$

PNF 必存在, 构造性证明, 略。

PNF 不唯一。

1.4 Rules of Inference

a **valid argument**: $(\bigwedge \text{premises}) \rightarrow \text{conclusion}$

fallacy: 谬误

1.4.1 Rules

Modus Ponens	肯定前件	$\begin{cases} x \\ x \rightarrow y \end{cases} \Rightarrow y$
Modus Tollens	否定后件	$\begin{cases} \neg y \\ x \rightarrow y \end{cases} \Rightarrow \neg x$
Hypothetical Syllogism	假言三段论	$\begin{cases} x \rightarrow y \\ y \rightarrow z \end{cases} \Rightarrow x \rightarrow z$
Disjunctive Syllogism	析取三段论	$\begin{cases} x \vee y \\ \neg x \end{cases} \Rightarrow y$
Addition		$x \Rightarrow (x \vee y)$
Simplification		$(x \wedge y) \Rightarrow x$
Conjunction		$\begin{cases} x \\ y \end{cases} \Rightarrow (x \wedge y)$
Resolution	消解	$\begin{cases} (x \vee y) \\ (\neg x \vee z) \end{cases} \Rightarrow (y \vee z)$

Constructive Dilemma	$\begin{cases} (x \rightarrow y) \wedge (u \rightarrow v) \\ x \vee u \end{cases} \implies (y \vee v)$
Destructive Dilemma	$\begin{cases} (x \rightarrow y) \wedge (u \rightarrow v) \\ (\neg y \vee \neg v) \end{cases} \implies (\neg x \vee \neg u)$
Universal Instantiation (UI)	$\forall x P(x) \implies P(u)$
Existential Instantiation (EI)	$\exists x P(x) \implies P(u) \text{ for some } u$
Universal Generalization (UG)	$P(u) \text{ for an arbitrary } u \implies \forall x P(x)$
Existential Generalization (EG)	$P(u) \text{ for some } u \implies \exists x P(x)$
Universal Modus Ponens	$\begin{cases} \forall x (P(x) \rightarrow Q(x)) \\ P(u) \end{cases} \implies Q(u)$
Universal Modus Tollens	$\begin{cases} \forall x (P(x) \rightarrow Q(x)) \\ \neg Q(u) \end{cases} \implies \neg P(u)$

prep.: using / from

e.g.

1. $\neg t$ premise 3.
2. $r \rightarrow t$ premise 2.
3. $\neg r$ modus tollens using 1, 2
4. $\neg r \vee \neg s$ addition from 3
5. $\neg(r \vee s)$ De Morgan's

1.4.2 Fallacies

- fallacy of affirming the conclusion:

$$\begin{cases} x \rightarrow y \\ y \end{cases} \not\Rightarrow x$$

- fallacy of denying the hypothesis:

$$\begin{cases} x \rightarrow y \\ \neg x \end{cases} \not\Rightarrow \neg y$$

1.5 Proof Methods

Direct Proof.

Conditional Proof (CP) 附加前提证明: to prove $A \implies (B \rightarrow C)$, use B as a premise. try to prove $(A \wedge B) \rightarrow C$.

Proof by Contraposition: use $\neg q \rightarrow \neg p$ to prove $p \rightarrow q$.

Proof by Contradiction 反证

Proof by Counterexample 找反例: use $\neg P(u)$ to prove $\neg \forall x P(x)$.

Existence Proof:

- a. Constructive: find a witness, that $P(u)$ is true.
- b. Nonconstructive

Exhaustive Proof / Proof by Exhaustion 穷举

Proof by Cases 分讨

Tips:

- Vacuous Proof: if p is false, then $p \rightarrow q$ is immediately true.
- Trivial Proof: if q is true, then $p \rightarrow q$ is always true.
- proof of equivalence:
 - a. to prove $p \leftrightarrow q$, proving $(p \rightarrow q) \wedge (q \rightarrow p)$.
 - b. to prove $p_1 \leftrightarrow p_2 \leftrightarrow \dots \leftrightarrow p_n$, proving $\bigwedge (p_i \rightarrow p_{i \bmod n+1})$.
- **WLOG**: without loss of generality.

第二章 Relations

Can't use emptyset for Cartesian Product.

2.1 Definitions

A binary relation R from A to B is a subset of $A \times B$: $R \subset A \times B$

If $(a, b) \in R$, then a is related to b . The **order** matters. Denoted as aRb or $(a, b) \in R$. Otherwise, $a \not R b$ or $(a, b) \notin R$.

The length of tuple is called degree. e.g. The degree of $A \times \cdots \times A$ is n . A_i is called the domains of R .

2.1.1 Representation

Order!

- **directed** bipartite graph
- **directed** graph for $R \subset A \times A$
- 01-matrix: $R : A \rightarrow B$, $M_{i,j} = [(a_i, b_j) \in R]$
 $M_{R \cap S} = M_R \wedge M_S$; $M_{R \cup S} = M_R \vee M_S$
 $M_{S \circ R} = M_S \odot M_R$ a.k.a. $M_{i,j} = \vee (M_{S_{i,k}} \wedge M_{R_{k,j}})$
 $M_{R^n} = M_R^n$

2.1.2 Special Relations

- 空关系 $\emptyset$
- 恒等关系 $I_A = \{(a, a) : \forall a\}$
- 全域关系 $U_A = \{(a, b) : \forall a \forall b\}$

2.1.3 Properties

- 自反性 Reflexive: $\forall x \in A, xRx$
Each node has a self-loop.
All elements on the main diagonal are 1.
- 反自反性 Irreflexive: $\forall x \in A, (x, x) \notin R$
Irreflexive is **not** the negation of reflexive !
 R can be neither reflexive nor irreflexive. R is both reflexive and irreflexive iff $A = \emptyset$ (however, Cartesian Product asks $A \neq \emptyset$ in $A \times A$, so?).
Each node must not have a self-loop.
All elements on the main diagonal are 0.
- 对称性 Symmetric: $R = R^{-1}$ (a.k.a. $\forall x, y \in A (x, y) \in R \iff (y, x) \in R$)
(等价地) All edges are bi-directed.
All are same across the main diagonal.
- 反对称性 Antisymmetric: $(x, y) \in R \wedge (y, x) \in R \implies x = y$
Antisymmetric is **not** "not symmetric" !
For any two different vertices, there are at most one directed edge link them.
All 1's are across from 0's.
- 非对称性 Asymmetric: $(x, y) \in R \implies (y, x) \notin R$
Completely no symmetric. Asymmetric $\iff$ Antisymmetric $\wedge$ Irreflexive.
No self-loops, and for any two different vertices, there are at most one directed edge link them.
All elements on the main diagonal are 0, and all 1's are across from 0's.
- 传递性 Transitive: $(x, y), (y, z) \in R \implies (x, z) \in R$

2.1.4 Operations

Inverse Relation: $R^{-1} := \{(b, a) : (a, b) \in R\}$.

e.g. $R = \{(a, b) : a < b\}$, then $R^{-1} = \{(a, b) : a > b\}$.

Note that if $R = \{(a, b) : p(a, b)\}$, R^{-1} is **not** $\{(a, b) : \neg p(a, b)\}$!

Properties of Inverse Relation:

- $(R^{-1})^{-1} = R$
- $(R \cap / \cup / - S)^{-1} = R^{-1} \cap / \cup / - S^{-1}$
- $R \subset S \iff R^{-1} \subset S^{-1}$
- $(R \circ S)^{-1} = S^{-1} \circ R^{-1}$

Composition: $S \circ R : A \xrightarrow{R} B \xrightarrow{S} C$

Power:

- $R^0 := I$
- $R^n := R^{n-1} \circ R$
- $R^{-n} := (R^{-1})^n$

Thm. R is transitive iff $R^n \subset R$ for all $n \in \mathbb{N}_+$

Thm. Composition of relations satisfies the associative law.

2.2 Closures of Relations

2.3 Equivalence Relations

2.4 Partial Orderings